1 Planning (15 marks)

Defining the problem (3 marks)

P1 p is the independent variable or vary p [1]
P2 q is the dependent variable or measure q [1]
P3 Keep (horizontal) velocity (v) constant [1]

Methods of data collection (5 marks)

M1 Labelled diagram of apparatus including method to vary p. [1]
M2 Method to determine position of ball on surface e.g. carbon paper/dye/video/sand. [1]
M3 Use ruler/caliper to measure p and/or q. [1]
M4 Method to ensure velocity is constant e.g. releasing ball from same height on a track/ spring loaded device or impulse device set to a constant value. [1]
M5 Method to ensure that the moved surface remains horizontal, e.g. spirit level/check height at different places. [1]

Method of analysis (2 marks)

A1 q^2 against p q against $\sqrt{p}$ p against q^2 $\sqrt{p}$ against q [1]

A2 $v = \sqrt{\frac{g \times \text{gradient}}{2}}$ $v = \text{gradient} \times \sqrt{\frac{g}{2}}$ $v = \sqrt{\frac{g}{2 \times \text{gradient}}}$ $v = \frac{1}{\text{gradient}} \times \sqrt{\frac{g}{2}}$ [1]

Allow valid logarithmic graph e.g. lg q against lg p, lg $2p$, lg $2p/g$ and valid calculation of v from y-intercept.

Safety considerations (1 mark)

S Reasoned method to prevent ball rolling on floor e.g. box below/storage box for balls/ sand box.
Reasoned method to prevent ball causing injury e.g. goggles/safety screen [1]

Additional detail (4 marks)

D Relevant points might include [4]
1 Method to ensure that velocity of ball is horizontal only when it reaches table, e.g. curved track.
2 Ensure that the ball leaves the table at 90°, e.g. set square/protractor on upper surface.
3 Detail on measuring q – location of landing position e.g. centre of crater/start of track.
4 Detail on determining location of zero position for p and q e.g. set square, plumb line.
5 Detail on method of determining position of ball e.g. slow motion playback including scale.
6 Take many readings of q for each p and average.
7 Straight line through the origin shows that p is proportional to q^2/relationship is valid – this mark may only be awarded when A1 is given.
8 Use of high density ball to minimise the effects of air resistance.

Do not allow vague computer methods.

[Total: 15]

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